

Descriptive statistics

Turning a pile of raw numbers into a few honest summary figures — using spreadsheet functions to calculate the average, median, count/frequency and standard deviation, and knowing which one actually answers the question.

What this key knowledge covers

This summary distils everything you need for **KK13 — Descriptive statistics** in Data Analysis. Work through the explanations, then use the worked good-vs-weak answers to see exactly how marks are won and lost. Tick off the revision checklist at the end before a SAC or exam.

Study summary

Why summarise at all?

Leo served 11 customers before recess. Their wait times (seconds), already sorted, were:

You *could* hand someone all 11 numbers. But nobody reads a spreadsheet to "get a feel" for it. **Descriptive statistics compress a dataset into a handful of figures that describe it** — its centre, its spread, its counts — so a human can understand it at a glance. The catch: **different summaries tell different stories**, and a careless choice can mislead.

A film with 40,000 reviews shows you **one star rating** — say 4.2. That single number is a descriptive statistic: it summarises 40,000 opinions so you can decide in a second. But it hides whether everyone agreed (all 4s) or the crowd was split (half 1s, half 5s averaging to 4.2).

Centre (average) and spread (standard deviation) together tell the real story. Leo's morning is exactly this problem in miniature.

Average (the mean)

The **average** — properly the **mean** — is the *balance point* of the data. Add every value, divide by how many there are. In a spreadsheet:

For Leo's 11 customers: $38+39+\dots+215 = 652$, divided by 11 = **59.3 seconds**.

Ten of Leo's eleven customers waited **under a minute**. Yet the mean says "59 seconds" — because the single 215-second catering order **drags the balance point to the right**. Every value tugs on the mean, so one **outlier** (an extreme value) can pull it well away from where most of the data actually sits. That's the whole reason the median exists.

Picture each wait time as a child sitting on a see-saw, at the position of their value. The mean is **where you'd put the pivot so the plank balances**. Move one child far out to the end (the 215 catering order) and the pivot has to slide a long way toward them to keep balance — even though nobody else moved. The next lab lets you feel exactly that.

Lab · the balance beam

Each ball is one of Leo's wait times sitting at its value along the beam. The **amber pivot** auto-slides to the **mean** (the balance point). The **blue ball** is the **median** (the positional middle). Drag the slider to grow that one catering order and watch what happens.

Median

The **median** is the *middle value* when the data is sorted in order. It cares only about **position**, not size — so an extreme value can't drag it around.

- **Odd count** (Leo's 11 values): the median is the single middle one — the 6th value = **44 s**.
- **Even count** (say 10 values): there's no single middle, so you average the **two middle values**. Pulled up ~15 s by one catering order. If Leo reports this as "typical wait", customers think the cart is slow.

Half waited less, half waited more. A **fairer** "typical wait" because the rare big order can't distort it.

When data is **skewed** or has **outliers**, the **median** is the better measure of "typical". When data is roughly **symmetric** with no extreme values, the **mean** uses all the data and is usually preferred. Saying *which* and *why* is where the marks are.

Count & frequency

Centre statistics answer "how big". **Count** and **frequency** answer "how many". They sound the same but examiners separate them:

How many entries in total (or how many meet one test). Leo had **40 orders** all morning.

How many fall into each category. Split those 40 orders by drink type → a **frequency table**.

Leo's morning, by drink

COUNT counts **numbers only** (ignores text). COUNTA counts **any non-empty cell** (text or numbers). COUNTIF counts cells that **meet a condition** — the engine of every frequency table. Mixing these up is a classic lost mark.

Count is the bouncer's clicker at the door — total people in. **Frequency** is the cloakroom tally — how many coats, how many bags, how many umbrellas. Same crowd, sorted into buckets.

Standard deviation

The mean tells you the centre. **Standard deviation (SD)** tells you the *spread* — on average, how far the values sit from the mean. A **small SD** means the data is tightly clustered (**consistent**); a **large SD** means it's widely scattered (**unpredictable**).

You will never hand-calculate SD in this course. You choose the function, then **interpret the result**. That's the examinable skill.

Same average, totally different experience

Two of Leo's helpers each served 8 customers. **Both averaged exactly 43 seconds**. The mean alone says they're identical — but they're not:

Two players both average dead-centre. One lands every dart in a tight cluster (low SD); the other sprays them around the board but they happen to average to the bullseye (high SD). The

average score is the same; the consistency is night and day. SD is the size of the cluster.

Lab · the spread-o-meter

Every dot is a wait time, floating around the fixed **mean line (43 s)**. Drag the slider to spread the dots out — the **± 1 SD band** grows and the SD readout climbs, **while the mean never moves**. That's the point: **spread and centre are independent**.

Choosing the right statistic

This is the part exams actually test. Match the **question being asked** to the **statistic that answers it honestly**.

Spreadsheet function reference

The exact functions for KK13, plus the close companions examiners expect you to recognise.

① Write the = sign — a formula without it is just text. ② Reference the **cell range** (B2:B12), never retype the numbers, so it updates live. ③ When you copy a formula, use **\$** to lock a cell you don't want shifting (e.g. =B2/\$B\$13 for a percentage of total).

P·E·L writing trainer

Short-answer marks come from **structure**, not length. Use **P·E·L**: make a **Point**, give a concrete **Example** from the data, then **Link** back to what was asked. Click a statistic to see a model paragraph.

VCAA-style practice questions

Five questions, **17 marks** total. Try each in your head or on paper first, then reveal the weak response, the strong response, the examiner's note and the marking guide. Self-mark honestly — your score builds in the dock.

Common traps & misconceptions

⚠ The "average is always best" trap

Quoting the mean for skewed data (Leo's 59 s) and calling it typical. → Spot the outlier/skew, switch to the median, and **say why**.

⚠ The population-vs-sample SD trap

Using STDEV.P on a sample (or vice versa) and giving no reason. → Measured everyone = population (.P); measured some = sample (.S). VCE data is usually a sample.

⚠ The COUNT-vs-COUNTA trap

Using COUNT on a column of drink names and getting **0** — because COUNT ignores text. → Text? Use COUNTA (any non-empty) or COUNTIF (by category).

⚠ The count-vs-frequency trap

Answering "how many of each drink" with a single total instead of a per-category breakdown. → "Of each" = **frequency** = a value per category, not one grand total.

⚠ The naked-number trap

Writing "SD = 51.8" with no units and no meaning. → Add units and interpret: "SD $\approx$ 52 s — very high, so wait times were inconsistent (driven by the catering order)."

⚠ The "more functions = more marks" trap

Listing every statistic instead of the one the question asked for. → Read the command word. **Calculate** wants a value; **justify/explain** wants the value *and* the reasoning.

How to answer exam questions

Read the **command word** first — it sets how much to write. *State/Identify* wants a word or phrase; *Describe* wants features; *Explain* wants reasons and links; *Justify/Evaluate* wants a judgement backed by evidence. Always pull in the **scenario detail** rather than answering in the abstract. The examples below contrast a weak response with a strong one for the same question.

Q1. Describe · 2 marks

A wellbeing survey records a daily mood score (1–10) for 200 students. Describe what the **standard deviation** of those scores would tell the analyst.

✗ A weak answer

It tells you the average mood of the students. (→ *Confuses standard deviation with the mean; describes the wrong statistic, so it earns 0.*)

✓ A strong answer

Standard deviation measures how **spread out** the mood scores are around the mean. A small standard deviation means most students scored close to the average (consistent wellbeing); a large one means scores varied widely, so the cohort's experience was much more mixed than the average alone suggests.

How the marks are awarded

- 1 mark — identifies it measures spread/variability around the mean.
- 1 mark — interprets what small vs large spread means for this dataset.

Q2. Explain · 3 marks

Explain why reporting only the **mean** of a dataset can be misleading, and name one other descriptive statistic that gives a fuller picture.

✗ A weak answer

The mean can be wrong sometimes so you should use other numbers too. (→ *Vague, names no statistic, gives no reason — 0-1.*)

✓ A strong answer

The mean is pulled toward extreme values (outliers), so a few very high or very low figures can make the average unrepresentative of a typical case. Reporting the **median** (the middle value) alongside it shows the central tendency without being distorted by outliers, and the **standard deviation** shows how spread the data is — together they describe the data far more honestly than the mean alone.

How the marks are awarded

- 1 mark — explains the mean is distorted by outliers/skew.
- 1 mark — names a suitable statistic (median, mode or standard deviation).
- 1 mark — links that statistic to a fuller/accurate picture.